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Original Article

Enhancing thermal conductivity of C/SiC composites containing heat transfer channels

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ABSTRACT

In this study, CNTs/SiC micro-pillars at controlled content ratios were introduced into C/SiC composites as heat transfer channels to improve the thermal conductivity in the thickness direction. The thermal conductivities and bending strengths before and after heat treatment at 1650 °C were investigated and the results were discussed. The theoretical calculations and finite element analyses confirmed that CNTs/SiC micro-pillars successfully worked as heat transfer channels. The theoretical thermal conductivity calculated by effective medium theory (EMT) model was 19.25 W/m·k and agreed-well with the experimental value. The measured thermal conductivity was estimated to 20.69 W/m·k and improved to 22.36 W/m·k after heat treatment. The latter was 3.56-fold higher than that of traditional C/SiC and attributed to increased grain growth during heat treatment. The optimal bending strength before heat treatment was recorded as 324.5 ± 23.74 MPa due to microstructure evolution caused by CNTs. After heat treatment, the bending strength improved by 138 % with ductile fracture mode attributed to ordered layer structure of PyC interphase and complex phase composition of the composites. These features benefited the abundant propagation of cracks and energy consumption. In sum, introduction of heat transfer channels into C/SiC composites provided a new way to improve the thermal conductivity in thickness direction of ceramic matrix composites.

1. Introduction

Continuous carbon fiber reinforced silicon carbide matrix composites (C/SiC) have attracted increasing attention in aerospace applications [1,2] owing to their low density, excellent mechanical properties, and superior oxidation resistance at high temperatures [3–6]. In addition, C/SiC composites have been extensively studied in combustion chambers [7] of scramjet engines, aerospace heat shields [8], and leading edges [9]. However, the poor capability of heat conduction, especially along through-thickness direction of C/SiC composites still limited their applications [10–14].

The thermal conductivity of ceramic matrix composites can be improved by introducing a second phase with high thermal conductivity (such as carbon nanotubes [15], SiC_{nw} [16], and diamonds [17,18]) into different components of C/SiC composites like interfaces [14], matrices [16–18], and coatings. Materials with great heat-conduction include carbon nanotubes (CNTs) characterized by high thermal conductivity, superior thermal stability, and extraordinary mechanical properties [19–21]. Chen et al. [22] produced vertically aligned CNTs on carbon fiber fabrics to form 3D hierarchical nanostructures. Polymer infiltration pyrolysis (PIP) method was then used to form C/SiC composites with improved through-thickness thermal conductivity of 16.8 W/m·k. Yang et al. [23] used no chemical-reaction method to embed CNTs within inter-bundles spaces of two adjacent carbon fiber sheets by vacuum infiltration to yield 3D-linked C_f-CNTs hybrid structures. The through-thickness thermal conductivities of CNT-SiCN composites modified by long and short CNTs enhanced to 5.6 W/m·k and 4.4 W/m·k, respectively. Feng et al. [24] employed vacuum filtration combined with chemical vapor infiltration method to fabricate SiC_f-CNTs/SiC composites containing few layers of SiC fiber cloths and buckypapers. Note that each layer was bonded to another by chemical vapor infiltration (CVI) SiC. The resulting thermal conductivity was estimated to 23.9 W/m·k, which was 2.9-fold higher than the traditional CVI-SiC/SiC composites. All these methods rely on orientation of CNTs but fail to fundamentally improve the thermal transfer ability due to the uncontrollable behavior of CNTs or other nanomaterials at the nanoscale [25–27]. On the other hand, construction of heat transfer channels to improve heat conduction properties in thickness direction is rarely reported. In our previous work, novel laser assisted-chemical vapor infiltration (LA-CVI) method combining the traditional CVI with laser processing was used to build mass transfer channels for CVI process [28–32]. LA-CVI method successfully strengthened the mechanical

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properties of C/SiC composites. Based on these data, we filled the mass transfer channels with CNTs to work as heat transfer channels.

To improve the thermal stability and release thermal residual stress, heat treatment is often used for processing in ceramic [33–35] or metal [36–38] matrix composites. Heat treatment is also used in performance evaluation of C/SiC composites at high temperatures. Yang et al. [39] studied the effects of heat treatment on CVI-Tyranno-SA/SiC composites and showed significant damage caused by sublimation of CVI-SiC matrix after heat treatment at 2273 K. Mei et al. [40] noticed enhancement in tensile strength of C/SiC composites composed of thin PyC interphase after heat treatment by 42 %. Guellali et al. [41] demonstrated that low-temperature PyC graphitizes easily than high-temperature PyC. Also, low-temperature PyC could transform into highly oriented pyrolytic graphite after heat treatment at very high temperatures.

In this work, CNTs/SiC micro-pillars inside C/SiC composites with different CNTs contents of micro-pillars were constructed as heat transfer channels. Finite element simulations and EMT model were employed to clarify the heat transport mechanisms and estimate the theoretical thermal conductivities. The experimental results showed substantially improved thermal conductivity (20.69 W/m·k), matching the calculated value. Besides, bending strengths of the composites were presented the disadvantageous for effect of CNTs/SiC micro-pillars. However, heat treatment significantly enhanced both thermal conductivities and bending strengths due to transformation of the crystalline structures.

2. Experimental

2.1. Materials

Multi-walled carbon nanotubes (MWCNTs) with outer diameters from 8 to 15 nm and lengths from 10 to 50 μm were supplied by Chengdu Organic Chemicals Co, Ltd. Deionized water was used as solvent for the preparation of MWCNTs-based suspensions. Triton X-100 (> 98 wt%) and tetramethyl ammonium hydroxide (aqueous solution 25 %) were provided by Sigma-Aldrich.

2.2. Preparation of CNT-C/SiC composites

C/SiC composites with internal CNTs/SiC micro-pillars were fabricated following three main steps. Semi-dense C/SiC composites (density 1.5 g/cm^3) were first fabricated by CVI method and Ti: sapphire laser system was employed to process holes with diameters around 0.5 mm. Details of the process could be found in reported literature [28]. Next, 0.2 g MWCNTs were dispersed in 500 ml deionized water, and 0.4 g triton X-100 and 0.6 g tetramethyl ammonium hydroxide were employed as dispersant and pH conditioning agent, respectively. After well-dispersing MWCNTs by ultrasonic process for 30 min, the as-obtained C/SiC composites were added to MWCNTs suspension followed by vacuum impregnation system for 30 min to fill the holes in C/SiC composites with MWCNTs. The resulting C/SiC composites were then dried in vacuum oven for 15 min as one cycle. Composites obtained for 5, 10, 15 and 20 cycles were named CS5#, CS10#, CS15# and CS20#, respectively. Note that the extremely lightweight of CNTs led to difficult confirmation of weight change of the samples after each filling cycle, especially CS5# and CS10#. As a result, the variate was set as number of filling times. For comparison, C/SiC composites with no micro-pillars (denoted as CS0#) were also prepared and tested. The as-obtained C/SiC composites were then densified by CVI method used during the first step. The detailed diagram of the preparation process is depicted in Fig. 1. For heat treatment, all samples were treated in vacuum environment at 1650 °C and preserved for 2 h.

2.3. Characterization

The density and open porosity of the composites were measured by the Archimedes method and the results are gathered in Table 1. Phase composition measurements were carried out by X-ray diffractometry (Rigaku D/max-2400, Japan). Three-point bending tests were obtained by an electro-mechanical universal testing system (Instron 5567, UK) at room temperature and loading rate of 0.5 mm/min. To this end, the prepared composites were processed into 40 mm × 5 mm × 3 mm specimens, which were used for each test. The microstructures of the composites were observed by scanning electron microscopy (SEM, FEI, Helios G4 CX, USA). The thermal diffusivity (α) and heat capacity (C_p) were measured by laser flash method (LFA 427, NETZSCH, Germany) at heating rate of 2 °C/min. The thermal conductivity (λ) was calculated according to the equation: $\lambda = \alpha \times C_p \times \rho$, where ρ is the specimen density and C_p is the specific heat capacity. The sample size was set to Φ12.7 mm × 3 mm. Analysis of SEM images was conducted by Image J software [42,43] (National Institutes of Health, USA). Finite element simulations were analyzed by the software ABAQUS (SIMULIA, France).

3. Results and discussion

3.1. Theoretical analysis of thermal conductivity

Theoretical calculations were first used to verify the effect of CNTs/SiC micro-pillars on thermal conductivities of C/SiC composites. According to the EMT proposed by Maxwell, Geiger and Hasselman [44], the interface thermal resistance caused by porosity and contact between different phases of the composites should play crucial role in thermal transport. Some models, such as Kopelman isotropic, Maxwell-Eucken and Bruggeman [45,46] were created to calculate the theoretical thermal conductivities of the composites. Here, two main steps were considered to calculate the thermal conductivity of C/SiC composites containing CNTs/SiC micro-pillars. The first consisted of thermal conductivity of CNTs/SiC micro-pillars and the second was based on thermal conductivities of the composites.

The thermal conductivity of CNTs could theoretically reach 3000 W/m·k for MWCNTs. However, this could directly degrade by one or two orders of magnitude due to twisting and aggregation. Feng et al. [14,24] used EMT theory to study the thermal conductivity of CNT-based composites. Considering CNTs/SiC micro-pillars as carbon nanotube-based composite with carbon nanotubes randomly dispersed in SiC matrix, Eq. (1) from EMT theory [47] could effectively describe the thermal conductivity of CNTs/SiC composites. Note that K_{mp} is the effective thermal conductivity of CNTs/SiC composite, K_{SiC} is thermal conductivity of SiC, K_x and K_z are respectively the thermal conductivities of carbon nanotubes along radial and axial direction [47], and f is volume fraction of CNTs. Singh et al. [48] found nanotube anisotropy to have low influence on the effective thermal conductivity of composites. Also, the thermal conductivity of CNTs was much higher than that of SiC. Eq. (1) could be simplified by Eq. (2). Here, thermal conductivity of 280 W/m·k for β-SiC was used in calculations [49]. Assuming decrease in thermal conductivity of aggregated CNTs to 600 W/m·k, at least 50 % volume fraction of CNTs was used in the preparation process. Therefore, the thermal conductivity of CNTs/SiC was calculated as K_{mp} of 569.98 W/m·k.

$$\frac{K_{mp}}{K_{SiC}} = \frac{3(K_x/K_{SiC}+1)+f\,[2(K_x/K_{SiC}-1)+(K_x/K_{SiC}+1)(K_z/K_{SiC}-1)]}{3(K_x/K_{SiC}+1)-2f\,(K_x/K_{SiC}-1)} \tag{1}$$

$$\frac{K_{mp}}{K_{SiC}} = \frac{3+fK_{cnt}/K_{SiC}}{3-2f} \tag{2}$$

As depicted in Fig. 2, the heat transfer mode of C/SiC composites

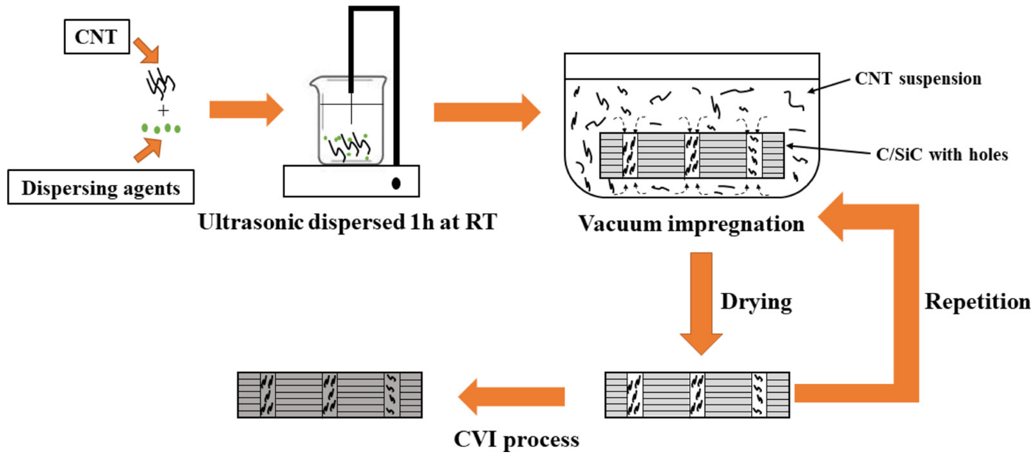

Fig. 1. Diagram showing the preparation process of the composites.

Table 1
Density and porosity of C/SiC composites.

Sample	Density (g· cm^{-3})	Porosity (%)
CS0#	2.23	8.15
CS5#	2.25	8.17
CS10#	2.21	8.13
CS15#	2.19	8.27
CS20#	2.16	7.69

containing CNTs/SiC micro-pillars could be described by the thermal resistance parallel model derived from the equivalent electrical resistance calculation theory. The model was used to study heterogeneous composites formed by superposition of different components (Fig. 2). Hence, the thermal conductivity of the composites could simply be given by Eq. (3), where K_e is the effective thermal conductivity of the composite, K_{cs} is thermal conductivity of C/SiC composites measured as 6.29 W/m·k, and v_{mp} is volume fraction of CNTs/SiC micro-pillars set to 2.3 % for micro-pillars with diameter of 0.5 mm, in which 15 micro-pillars were introduced into each test sample. The obtained theoretical thermal conductivity of C/SiC composites was 19.25 W/m·k.

$$K_e = (1 - v_{mp})K_{C/S} + v_{mp}K_{mp} \quad (3)$$

Finite element simulations were employed to analyze the effects of orientated micro-pillars as heat transfer channels. As shown in Fig. 2a, standard testing model of thermal conductivity mentioned in Section 2.3 was created in ABAQUS software. A continuous heating source of 1500 °C was applied on the top surface C/SiC model while ambient temperature was set to 20 °C. The measured thermal conductivities of C/SiC composites and those of CNTs/SiC micro-pillar calculated by theoretical equations were obtained. According to literatures, heat capacities of CNT [50,51], β-SiC [52] and C/SiC [53] would increase with temperature. We set heat capacities of those as average value 600 J· kg^{-1}· K^{-1}, 950 J· kg^{-1}· K^{-1} and 1315 J· kg^{-1}· K^{-1}, respectively. In addition, based on rule of mixture of composite, the heat capacity of the CNTs/SiC pillar ranged from 670 J· kg^{-1}· K^{-1} to 775 J· kg^{-1}· K^{-1} (changing with CNT content). The simulations were based on two conditions. First, heat radiation and heat convection were ignored and heat transfer was considered. Second, scattering contributions were ignored [23]. In Fig. 2b and c, the cross-section of the models after 5 s, 10 s and 15 s heating processing displayed C/SiC composites with and without micro-pillars at different heat transfer modes. As shown in Fig. 2b, CNTs/SiC micro-pillars surely acted as heat transfer channels inside the composites due to elevated thermal conductivity of CNTs/SiC micro-pillars. In addition, CNT/SiC micro-pillars could act as additional heat sources to accelerate heat diffusing inside composites in all directions due to their higher temperatures than side C/SiC areas. This contributed to faster and uniform temperature distributions when compared to traditional

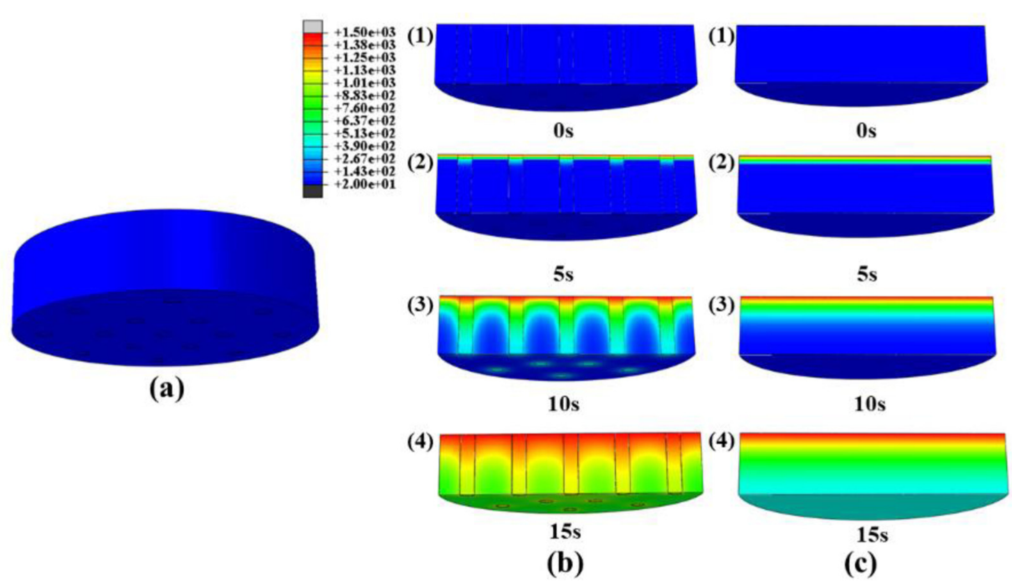

Fig. 2. Finite element simulations of C/SiC composites with and without heat transfer channels.

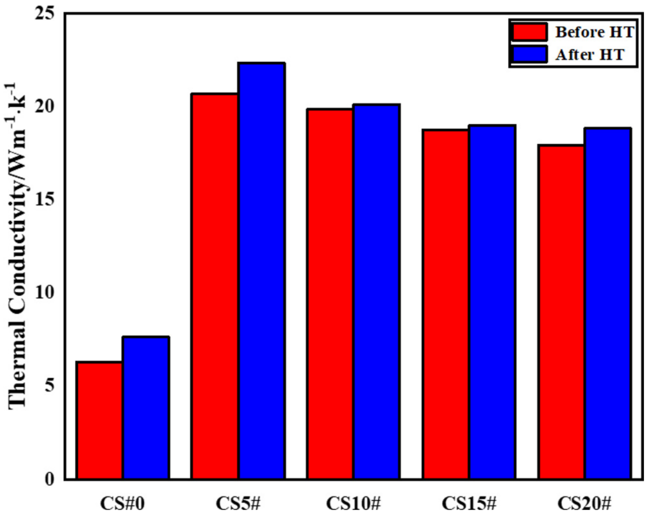

Fig. 3. Thermal conductivities of C/SiC composites prepared with CNTs/SiC micro-pillars before and after heat treatment.

C/SiC composites. As a consequence, the introduction of CNTs/SiC micro-pillars could theoretically improve the thermal conductivities along the thickness and in-plane direction.

3.2. Thermal conductivity of C/SiC composites containing CNTs/SiC micro-pillars

Fig. 3 exhibits the thermal conductivities of C/SiC composites with CNTs/SiC micro-pillars. Compared to traditional materials, C/SiC composites containing CNTs/SiC micro-pillars illustrated outstanding thermal conductivities around 19.32 W/m·k. As CNTs filling cycles increased, thermal conductivities decreased from 20.69 W/m·k to 17.94 W/m·k. The content ratio of CNTs in the micro-pillars obtained by calculations mentioned in Section 3.1 was estimated to 50 %. Depending on the filling time, content ratio of CNTs could vary from 50 % to 80 %. Therefore, the theoretical values of thermal conductivity changed from 19.25 W/m·k to 21.32 W/m·k. Hence, the experimental values greatly agreed with the theory but showed the opposite trend as a function of CNTs filling time (Fig. 4). This could be explained by density and porosity of the composites. The theoretical Eqs. (1)–(3) described the relationship between different components of the

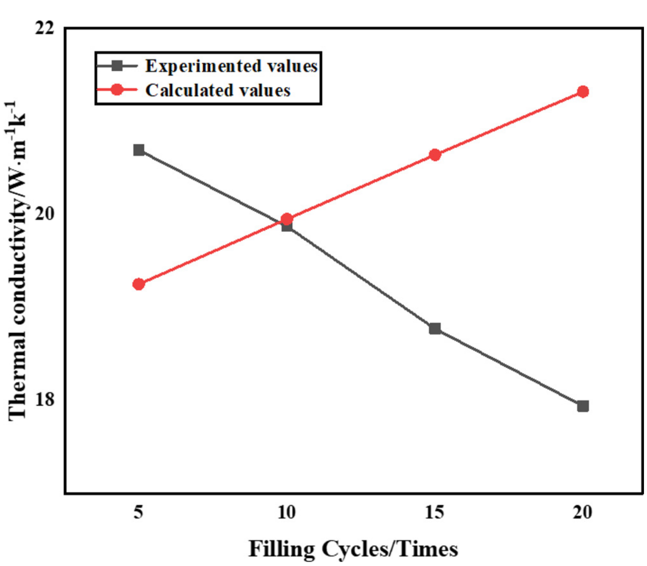

Fig. 4. Calculated and experimental conductivities of C/SiC composites containing micro-pillars.

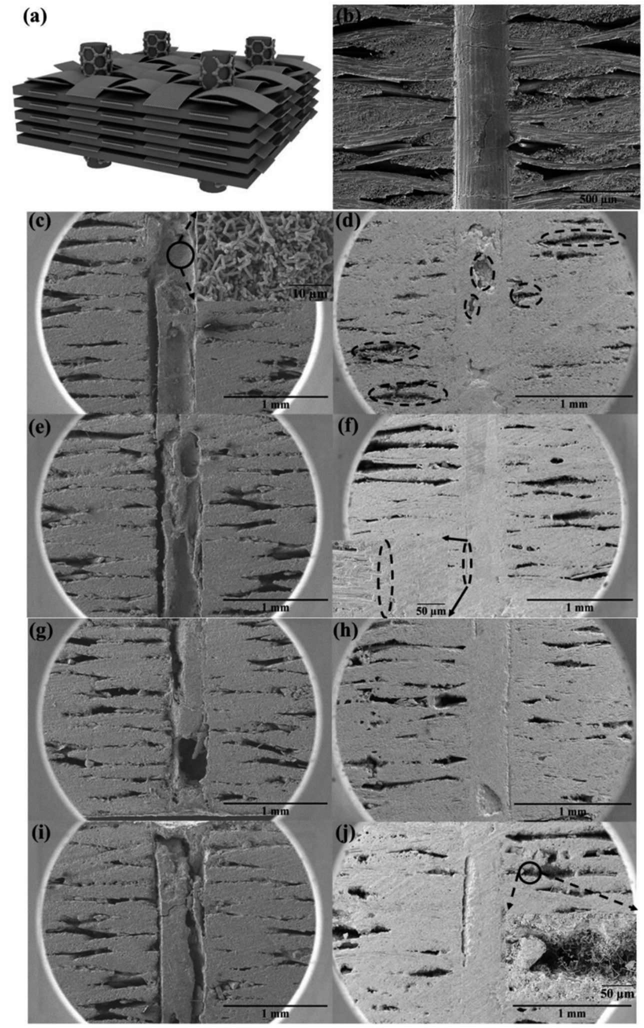

Fig. 5. CNTs/SiC micro-pillars of C/SiC composites: a) structure of the composites, b) C/SiC composites with holes, c) CS5#, e) CS10#, g) CS15#, i) CS20# with density of 1.8 g/cm^3, d) CS5#, f) CS10#, h) CS15#, and j) CS20# with density of 2.1 g/cm^3.

composites but failed to quantify the influence of contact thermal resistance caused by changes in components and other influential factors when deduced and created. Hence, CNTs and micro-pillars, as well as components with higher thermal conductivities all play crucial roles in theoretical values of thermal conductivity. However, the porosity and other microstructural changes should be considered as influential factors for the experimental values of thermal conductivity. Scanning electron microscope (SEM) (Fig. 5c-j) exhibits the microstructure evolution of the composites. In Fig. 5-d, f, h and j, CNTs/SiC micro-pillars inside C/SiC composites were formed when composites density reached to 2.1 g/cm^3. Upon filing of CNTs with 5 cycles, large amounts of CNTs were coated with SiC on the wall of hole (magnified image in Fig. 5-c). In addition, various-sized cavities were infiltrated with SiC during CVI process, indicating different content ratios of CNTs and SiC in different micro-pillars. Thermal conductivity of CS5# reached 20.69 W/m·k, and increased to 22.36 W/m·k after heat treatment. According to the literature [14,15,24], interface thermal conductance generated by CNTs would greatly degrade the thermal conductivity of CNTs composites. Singh et al. [48] described the influence of the interface on thermal conductivity of the composites through continuum mechanics, where longer CNTs had more effect than shorter CNTs on the interface resistance. Increase in CNTs content would bring more interfaces to SiC and generate more contact for thermal resistance and large quantities of

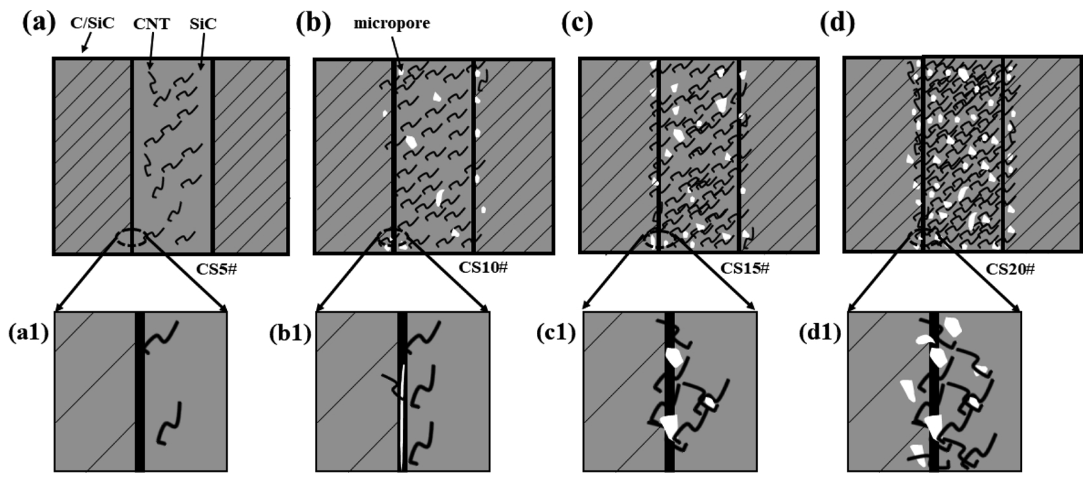

Fig. 6. Microstructures of composites containing micro-pillars.

unavoidable micropore structures. These features should reduce the thermal conductivity, as shown in Fig. 6b–d.

After heat treatment, the thermal conductivities of C/SiC composites slightly increased with average value of 20.08 W/m·k. According to Smith et al. [54], heat conduction in solid non-metallic materials should be dominated by phonons caused by lattice vibrations. At grain boundaries, crystallite orientation would change but some disordered crystalline structures should remain, changing phonons transfer path and direction. Lee et al. [55] and Jang et al. [56] studied the influence of mean grain size on thermal conductivity of sintered SiC ceramics. They showed that mean grain size increased with temperature, beneficial for improved thermal diffusivity due to scattering of phonons by grain boundaries, leading to size-dependence. Grains would grow at high temperatures to generate high purity grains with integrated crystalline structure, contributing to elevated phonons transport efficiency between grain boundaries [57]. Fig. 7 displays the X-ray diffraction (XRD) patterns of CS5# before and after heat treatment. Before heat treatment, CS5# consisted of PyC, α-SiC, β-SiC, and SiO. Note that SiO represented the ablation product generated by laser processing. The transformation of SiC grains from β-SiC to α-SiC and coarsing of β-SiC grains occurred at 1650 °C [40], as demonstrated by narrow and high intensities β-SiC and α-SiC diffraction peaks. Larger and integrated grains reduced grain boundaries and defects, providing longer mean for free path of phonons that led to further improved thermal conductivities. Fig. 8 displays data from recent studies dealing with thermal conductivity of fiber-reinforced ceramic matrix composites, such as C/SiC, SiC/SiC and C/SiCN modified by CNTs, diamond, SiC_{nw}, and Ti_3SiC_2. Composites with high densities showed better thermal conductivities. Hence, introduction of heat transfer channels could effectively enhance the thermal conductivities of composites.

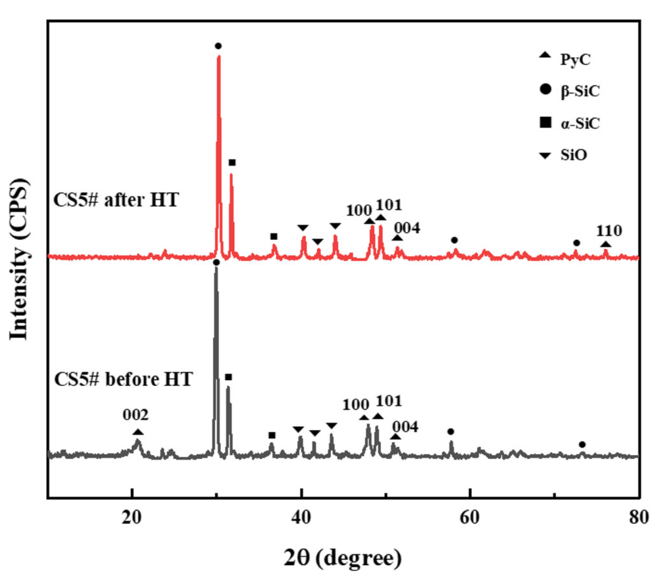

Fig. 7. XRD patterns of CS5# and CS5# after heat treatment.

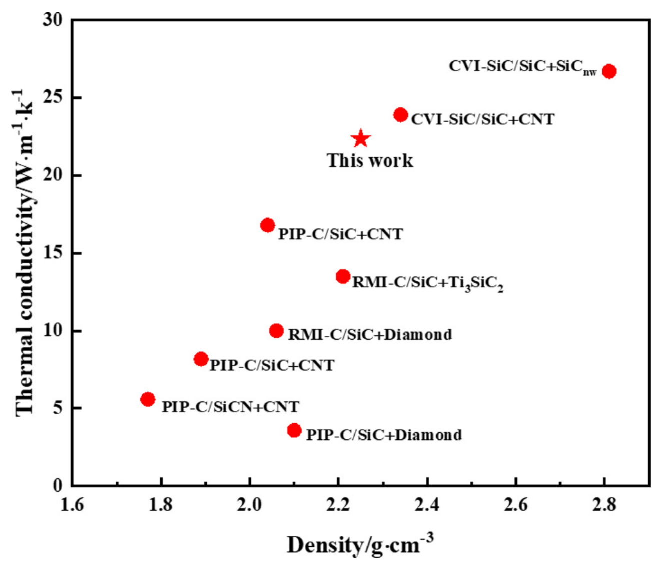

Fig. 8. Recently reported thermal conductivities of fiber-reinforced ceramic matrix composites [15–17,22–24,58,59].

3.3. Bending strength

Liu et al. [28] used LA-CVI method to improve the mechanical properties of C/SiC composites with appropriate holes by forming "dense band" with pinning effect. In this work, heat transfer channels introduced into C/SiC composites could reduce the mechanical properties by cutting off continuous fibers and forming holes filled with CNTs. The bending strengths of the composites before and after heat treatment are gathered in Fig. 9. As CNTs content in micro-pillars rose, bending strength of the composites increased and then decreased.

Table 2 lists the bending strength and modulus of composites before and after heat treatment. Before heat treatment, the composites showed brittle fracture with smooth fracture surfaces and extremely short fibers pulling out (Fig. 10a–b). Obviously, CNTs/SiC micro-pillars played negative role in bending strength and should be considered as flaws when compared to CS0#. However, CS15# slightly enhanced the

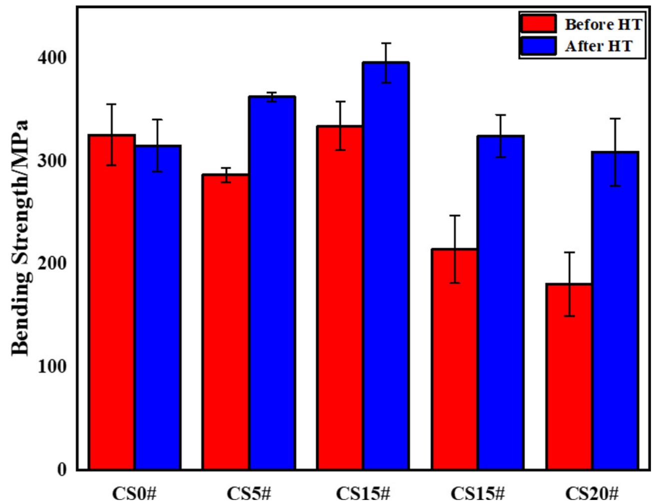

Fig. 9. Bending strength of C/SiC composites containing CNTs/SiC micro-pillars before and after heat treatment.

Table 2
Bending strength and modulus of C/SiC composites containing CNTs/SiC micro-pillars before and after heat treatment.

Sample	Bending Strength (MPa)		Bending Modulus (GPa)	
	Before HT	After HT	Before HT	After HT
CS0#	325.86 ± 29.68	315.31 ± 25.33	55.31 ± 2.16	31.27 ± 2.44
CS5#	286.91 ± 6.89	362.69 ± 4.43	28.86 ± 0.11	29.84 ± 0.67
CS10#	324.5 ± 23.74	395.92 ± 19.01	28.26 ± 8.30	26.63 ± 2.27
CS15#	214.5 ± 32.87	324.64 ± 20.88	42.14 ± 3.43	28.07 ± 0.17
CS20#	180.56 ± 31.03	308.97 ± 32.74	35.87 ± 1.37	31.32 ± 0.98

bending strength due to evolution in porosity and microstructure of the composites. In Fig. 5d, f, h and j, more flaws and cavities remained as filling cycles of CNTs increased, and more large-size pores formed around the micro-pillars. CNTs would clearly block mass transfer channels of CVI, resulting in advanced "bottleneck effect" [28]. In this work, the unique hole structure of the composites rendered the mercury intrusion useless for analysis of pore size distribution. Hence, Image J software was used to analyze the pore structures. Labeled pores and cavities areas in SEM images of Fig. 5d were calculated by Image J software and the results are listed in Table 3. Similar to changes in density trends (Table 1), the pores areas in each SEM picture showed "bottleneck effect" caused by CNTs. Another significant evidence could be noted in Fig. 5f and magnified image. CNTs partly obstructed the mass transfer channels and apparent cavity remained. On the other hand, combination between micro-pillars and C/SiC composites influenced the bending strength. First, the tight combination generated excess stress inside the composites, leading to fracture at the micro-pillar positions ahead of the matrix (Figs. 5d and 6 a1). Second, "bottleneck effect" caused by CNTs remarkably influenced the pore structure and material composition at the interface of micro-pillars and C/SiC (Figs. 5f, h, j, 6 b1, c1, and d1). Another evidence could be seen in the magnified image of Fig. 10d. CNTs coated with thin SiC exited in interlayers of the composites, which could be advantageous for interface debonding [60]. As mentioned in Section 3.2, CNTs would not only generate numerous pores or cavities inside micro-pillars but also at the interfaces. Compared to CS#10, the pores presented in/around the interfaces of CS#15 and CS#20 became local defects that may degrade the bending strength. By comparison, CS10# exhibited "suitable connection" with C/SiC material between micro-pillars and C/SiC (Fig. 5f, magnified image). This would release stress and become solid medium to effectively transmit the bending load and blunt crack-tip in pillars during bending loads [59].

Table 3
Image J analysis results of SEM photographs.

Sample	Pores area (mm^2)
CS5#	0.25
CS10#	0.24
CS15#	0.25
CS20#	0.30

After heat treatment (Fig. 9), the composites showed the same trend

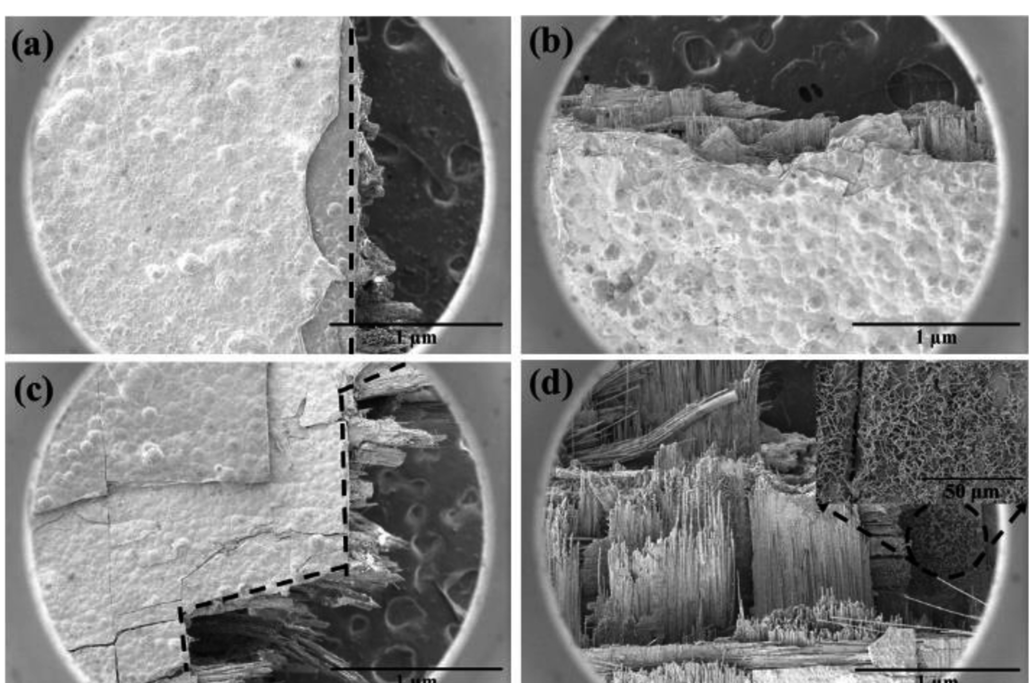

Fig. 10. SEM images of C/SiC composites containing CNTs/SiC micro-pillars: (a, b) before and (c, d) after heat treatment.

as before heat treatment but with significantly improved bending strengths. Meanwhile, fracture mode transformed from brittle to ductile fracture (Fig. 10c–d). Apparently, longer fibers pull-outs and step crack propagation paths led to enhanced mechanical properties. In Table 2, CS10# showed improved bending strength of 395.92 ± 19.01 MPa with bending strength after heat treatment improvement of 138%. Besides, Young's modulus after heat treatment was lower than that obtained before heat treatment (Table 2). Although failed to reach our best bending strength of 3D C/SiC composites (441 MPa [4]), C/SiC composites fabricated in this work should meet the application requirements of engineering.

The coarsening of SiC grains [39] is an adverse factor of strengthening and toughening, attributed to self-cracking after heat treatment [40]. However, heat treatment should contribute to graphitization of PyC interphase, CNTs, and carbon fibers. As showed in XRD, the diffraction peak of C (002) became wider and lower in intensity while that of C (110) became narrower and sharper with heat treatment. Meanwhile, the diffraction peaks of C (100) and (101) preserved similar intensities, indicating evolution to more complex C phase structures. After treatment at 1650 °C, C phases of PyC, CNTs and carbon fibers formed ordered layer structures to degrade interfacial sliding resistance and relieve the thermal residual stress in C/SiC and micro-pillars. According to mechanisms of interphase strengthening, composites after heat treatment would present enhanced toughness attributed to PyC interphase structure evolution. Ordered layer structure of PyC formed after heat treatment will decrease the friction coefficient between the matrix and fibers, beneficial for fiber breakage inside the matrix and pull out. Furthermore, such structures could produce crack deflection and even split. On the other hand, grains of β-SiC transformed to α-SiC could lead to more complex phase compositions (Section 3.2), which could consume more energy and lead to abundant cracks propagation mode [34,40]. Hence, abundant cracks and lamellar matrix destruction were observed (Fig. 10c).

The thermal conductivities and bending strengths showed potential for further improvement of thermal and bending strength of C/SiC composites containing CNTs/SiC micro-pillars. For example, higher density C/SiC composites could be employed to process holes with less remained porosity to prevent the influence of porosity on the composites. On the other hand, reaction melting infiltration (RMI) could also be effective in producing C/SiC composites with fewer pores. In addition, SWCNTs and DWCNTs with better thermal and mechanical properties could be used to fabricate micro-pillars according to theoretical equations. Above all, the proposed methodology may be applied to other ceramic matrix composites such as SiC/SiC and C/Si_3N_4 to extend the application fields of these composites.

4. Conclusions

CNT/SiC micro-pillars were successfully introduced into C/SiC composites by CVI method at controlled content ratio of CNTs and SiC of micro-pillars. Thermal conductivities and bending strengths of the composites before and after heat treatment at 1650 °C were studied and the results were discussed. The following remarks could be drawn:

Analysis of heat conduction properties by finite element modeling and theoretical calculations indicated that CNTs/SiC micro-pillars could work as heat transfer channels in C/SiC composites. Also, composites containing CNTs/SiC micro-pillars showed more uniform temperature distributions than traditional C/SiC composites. The theoretical thermal conductivity calculated by EMT model was 19.25 W/m·k.

The thermal conductivities before and after heat treatment showed C/SiC composites containing CNTs/SiC micro-pillars with outstanding thermal conductivities reaching maximum of 20.69 W/m·k. After heat treatment, thermal conductivity improved to 22.36 W/m·k due to integrated crystalline structure formed at high temperatures. The experimental values matched well the calculated data.

The bending strength before heat treatment revealed the negative influence of CNTs/SiC micro-pillars on bending strength. Large amounts of CNTs would block the mass transfer channels and generate pores structures in/around the micro-pillars. The microstructure evolution at interfaces between micro-pillars and C/SiC also impacted the bending strengths. Nevertheless, the bending strength enhanced to 395.92 ± 19.01 MPa after heat treatment due to the ordered layer structure of PyC interphase and more complex phase constitution assigned to energy consumption and abundant cracks propagation path.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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